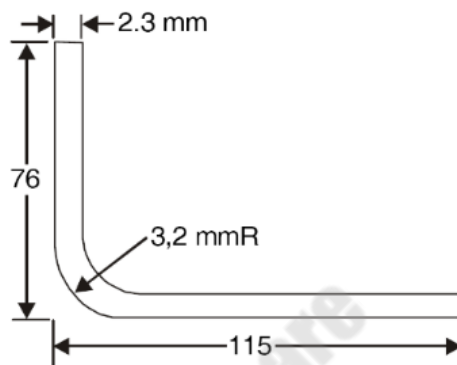


Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q. 1 Solve Any Two of the following.		
A) What are the desirable properties of materials in sheet metal forming?	(CO2)	6
B) Enlist the materials used in automotive sheet metal part manufacturing and explain any two of them.	(CO2)	6
C) Write the advantages of products made from sheet metal.	(CO2)	6
Q.2 Solve Any Two of the following.		
A) Explain the concept of Piercing and blanking with neat sketches.	(CO1)	6
B) List out different sheet metal operations and explain perforating and lancing operation in details	(CO2)	6
C) Determine the punch and blank sizes for blanking a circular disc of 20 mm diameter from a C20 steel sheet whose thickness is 1.5 mm. Shear strength of annealed C20 steel is 294 MPa. Also determine the required punching force.	(CO2)	6
Q. 3 Solve Any Two of the following.		
A) Distinguish between bending and drawing in sheet metal operation.	(CO3)	6
B) Describe with neat sketches the V-Bending and Wipe (Edge) bending operations.	(CO3)	6
C) Estimate the length of sheet necessary to manufacture the component as shown in the fig. The thickness of the sheet is 2.3mm and it is made of C10 material.	(CO2)	6



(All dimensions are in mm)

Q.4 Solve Any Two of the following.

- A) Explain Progressive die with neat sketch. (CO4) 6
- B) Explain the process of spinning for manufacturing of Axi-Symmetric components. (CO1) 6
- C) Compare the Mechanical press with Hydraulic press used in sheet metal stating their advantages and disadvantages. (CO5) 6

Q. 5 Solve Any Two of the following.

- A) Give any two applications of each of the following operations. (CO3) 6
1. Perforating 2. Lancing 3. Notching and 4. Bending operations.
- B) Suggest which sheet metal operations are used to manufacture the following parts. (CO3) 6
1. Cup 2. Washbasin 3. Part which contains array of holes 4. Coin
- C) Describe in brief any one case study of sheet metal manufacturing of a product highlighting the following. (CO1) 6
- a) Generic name of the product b) Material Used c) Types of Sheet Metal operations involved d) Sequence of Operations.

***** End of paper*****

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(Level/CO) Marks

Q. 1 Solve Any Two of the following.

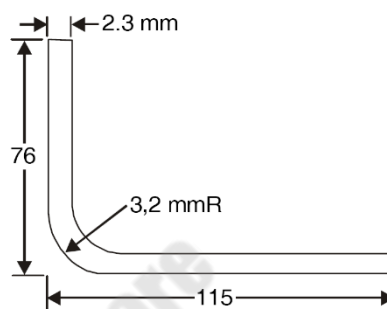
- | | | |
|---|-------|---|
| A) State & explain the advantages of manufacturing the products from Sheet Metal. | (CO2) | 6 |
| B) Enlist any six important properties of sheet metal material and explain in short, any two of them. | (CO2) | 6 |
| C) Write short note on any two of the following commonly used sheet metal materials
a) Stainless Steel b) Aluminum c) Copper | (CO2) | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|--|-------|---|
| A) Explain the difference between punching and blanking operation with neat sketches and suitable examples. | (CO3) | 6 |
| B) Explain the mechanics of shearing with neat sketches. | (CO3) | 6 |
| C) Estimate the punching force required to punch the hole of 20mm diameter in a plate having the thickness of 2 mm. The maximum shear strength of the material is 400 MPa. | (CO3) | 6 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|-------|---|
| A) Estimate the length of sheet necessary to manufacture the component as shown in the fig.1. The thickness of the sheet is 2.3mm and it is made of C10 material. | (CO3) | 6 |
|---|-------|---|



(All Dimensions are in mm)

- B) Write short note on any two of the following. a) Deep Drawing Operation (CO3) 6
b) Ironing Operation c) Defects in Drawn parts
- C) Explain with neat sketches, V-Bending and Wipe Bending along with suitable examples. (CO3) 6

Q.4 Solve Any Two of the following.

- A) Explain working of progressive Die giving suitable example with neat sketch (CO4) 6
- B) Explain working of Compound Die giving suitable example with neat sketch. (CO4) 6
- C) Classify and explain types of presses with their applications. (CO5) 6

Q. 5 Solve Any Two of the following.

- A) Describe any one case study of sheet metal manufacturing highlighting the following. (CO1) 6
a) Generic name of the product with uses b) Material Used c) Types of Sheet Metal operations involved d) Sequence of Operations with brief description.
- B) State the importance of center of pressure and how it is determined? Explain with suitable example. (CO1) 6
- C) A washer having the outside diameter of 30mm and inside diameter of 20mm is to be made. Determine the die and punch sizes required for punch & dies used in punching and blanking operations. Thickness of the sheet is 2.5mm and shear strength of the material is 450 MPa. (CO1) 6

***** End *****